

HCI-Chapter 1: Introduction

1. Overview (课程概览)

本章主要内容：

1. Concept of HCI (HCI概念)
 2. Definition of HCI (HCI定义)
 3. Importance of HCI (HCI重要性)
 4. Scope of HCI (HCI范围)
 5. Evolution of HCI (HCI演化)
 6. History and Trends (历史与发展趋势)
 7. Usability (可用性)
-

2. Introduction (引言)

Interactive Products (交互式产品)

现代社会大量产品都需要用户进行交互：

Examples:

- Smartphone
- Laptop
- Tablet
- TV Remote
- Coffee Machine
- Ticket Machine
- Game Controller

核心问题：

How many are actually easy, effortless and enjoyable to use?

(这些产品中有多少是真正容易使用、毫不费力且令人愉快的？)

Why HCI Matters

很多产品：

✓ Functionally correct

但：

✗ Difficult to use

如果产品按照：

- Programmers' preferences
- Designers' preferences
- Engineers' preferences

设计，而不是：

- Users' needs
- Users' capabilities

那么即使功能正常，也会导致：

- Frustration (挫败感)
- High Learning Curve (学习成本高)
- Poor User Experience (体验差)

3. Key Concept

Human-Computer Interaction (HCI)

Definition:

A multidisciplinary field encompassing the study of how people interact with computers and technologies, including research, theory, and design principles.

中文：

研究人类如何与计算机及各种技术进行交互，并通过研究、理论和设计原则改善这种交互的跨学科领域。

关键词：

- Human
- Computer
- Interaction
- Research
- Theory
- Design Principles

特点：

HCI is Research-Oriented.

(HCI偏向研究)

Interaction Design (IxD)

Definition:

A design discipline that applies HCI principles to create interactive products and experiences.

中文:

将HCI原理应用到产品设计中的设计学科。

特点:

Interaction Design is Design-Oriented.

(IxD偏向设计)

HCI vs IxD

HCI	IxD
Research-Oriented	Design-Oriented
Study interactions	Design interactions
Why interactions happen	How interactions should happen
Theory	Practice

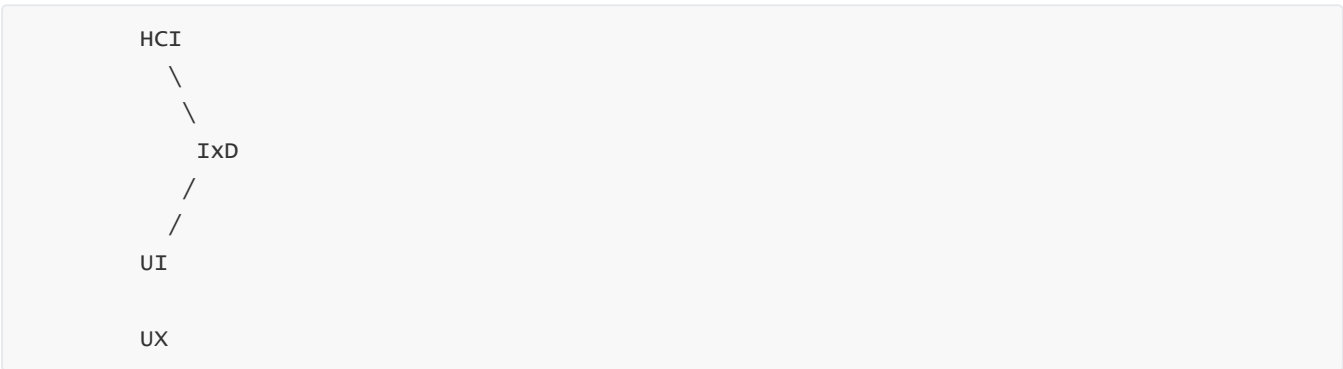
考试常考:

HCI studies interaction.

IxD designs interaction.

4. Relationship Between HCI, IxD, UI and UX

课件中的维恩图:



Interaction Design (IxD)

位于:

- Human-Computer Interaction (HCI)

- User Interface Design (UI)
- User Experience Design (UX)

之间。

理解：

HCI 提供理论

↓

IxD 提供设计方法

↓

UI 实现界面

↓

UX 体现体验

5. Components of HCI

HCI 由四个核心部分组成：

1. Human

人

可以是：

- Individual User
- Group of Users
- Organization

2. Computer

计算机

不仅仅是 PC

包括：

- Desktop Computer
 - Large-scale Computer System
 - Pocket PC
 - Embedded System
 - Search Engine
 - Word Processor
-

3. User Interface (UI)

Definition:

Parts of the computer that the user contacts with.

中文:

用户接触到的系统部分。

例如:

- Buttons
- Menus
- Screen
- Keyboard

4. Interaction

Definition:

A dialogue involving feedback and control between user and system.

中文:

用户与系统之间持续的反馈和控制过程。

Example:

User ↓ Input

System ↓ Output

User receives feedback

6. Why is HCI Important?

HCI bridges:

Human ↔ Technology

设计时需要考虑:

Efficiency (效率)

完成任务的速度

Effectiveness (有效性)

能否完成目标

Emotion (情感)

用户感受如何

Economy (经济性)

成本是否合理

Health (健康)

是否影响身体健康

Safety (安全)

是否容易出错

7. Who is Involved in Interaction Design?

Interaction Design 需要跨学科团队：

Examples:

- Engineers
- Designers
- Programmers
- Psychologists
- Anthropologists
- Sociologists
- Marketing Specialists
- Product Managers
- Artists

目的：

Provide good user experiences.

满足用户需求。

8. What to Design?

设计时需要考虑：

Who are the users?

用户是谁？

What activities are being carried out?

用户要做什么？

Where is the interaction taking place?

用户在哪里使用？

目标：

Optimize interactions.

使产品符合用户活动与需求。

9. Usability（可用性）

Definition

Usability:

Ensuring products are:

- Easy to learn
- Effective to use
- Enjoyable to use

从用户角度衡量系统是否好用。

10. Dimensions of Usability

课件重点

共六个维度

1. Learnability

容易学习吗？

2. Efficiency

学会后使用是否高效？

3. Effectiveness

能否帮助用户达成目标？

4. Safety / Error Tolerance

错误是否少？

错误是否容易恢复？

5. Satisfaction

用户是否满意？

6. Memorability

是否容易记住？

11. Different Users, Different Needs

Novice Users

关注：

Learnability

Infrequent Users

关注：

Memorability

Expert Users

关注：

Efficiency

12. Usability Goals

Definition:

Specific, measurable objectives set during the design process.

(设计过程中设定的可测量可用性目标)

Examples:

Effectiveness

≥90% users successfully register course

Efficiency

Complete registration within 3 minutes

Safety

Detect invalid course selection

Learnability

No instructor assistance required

Memorability

No relearning required

Satisfaction

User rating ≥4/5

13. User Experience (UX)

Definition

UX refers to:

The singular and accumulated experiences that occur for users as a consequence of interacting with a product.

中文:

用户在与产品交互过程中产生的整体体验。

重点:

How people feel.

用户的感受。

14. UX Goals

Desirable Aspects

正向体验：

- Satisfying
- Enjoyable
- Engaging
- Pleasurable
- Exciting
- Entertaining
- Helpful
- Motivating
- Fun
- Rewarding
- Emotionally Fulfilling

Undesirable Aspects

负向体验：

- Boring
- Frustrating
- Annoying
- Unpleasant
- Patronizing
- Childish
- Gimmicky

15. Usability vs User Experience

Usability

Core Question:

Can they do it?

关注：

- Functionality
- Efficiency
- Error Rate

客观 (Objective)

User Experience

Core Question:

Did they enjoy doing it?

关注:

- Emotion
- Satisfaction
- Feeling
- Aesthetics

主观 (Subjective)

16. Evolution of HCI

Early Interaction

输入:

- Wires
- Punch Cards
- Paper Tape
- Keyboard

输出:

- Lights
- Paper
- Teletype

特点:

- Expensive
 - Difficult to use
 - Not for mass market
-

Early Computers

ENIAC (1943)

- 15m × 9m
- Hydrogen bomb calculations

Mark I (1944)

Stretch (1961)

17. Modern Devices

Input Technologies

- Touchscreen
 - Multi-touch
 - Keyboard
 - Mouse
 - Stylus
 - Voice Input
 - Computer Vision
 - Gesture Sensors
 - Wearables
-

Output Technologies

- High-resolution Displays
 - Interactive Screens
 - Spatial Audio
 - Haptic Feedback
 - Smartwatches
 - AR/VR/MR
-

18. Future of HCI

Emerging Technologies:

- Conversational AI
- Eye Tracking
- Brain-Computer Interface (BCI)

- Gesture + Voice + Vision
- Adaptive Displays
- AI-generated Content
- Robotic Interaction
- Autonomous AI Agents

核心观点:

New interaction technologies are changing how humans communicate, perceive and collaborate with computers.

19. Early Concepts of HCI

最初的 User-Friendly 概念:

只是:

- Tidy Screens
- Better Appearance

而不是:

- Interaction Design
- User-Centered Design

研究重点逐渐转向:

- Human Capabilities
 - Human Limitations
 - Psychological Factors
-

20. Chapter Summary (考试必背)

HCI

Study of interaction between humans and computers.

IxD

Design of interaction.

Components of HCI

- Human
 - Computer
 - User Interface
 - Interaction
-

Importance of HCI

- Efficiency
 - Effectiveness
 - Emotion
 - Economy
 - Health
 - Safety
-

Usability Dimensions

- Learnability
 - Efficiency
 - Effectiveness
 - Safety
 - Satisfaction
 - Memorability
-

UX Core Question

Did they enjoy doing it?

Usability Core Question

Can they do it?

Future HCI

- Conversational AI
- BCI
- Eye Tracking
- AR/VR

- Autonomous Agents